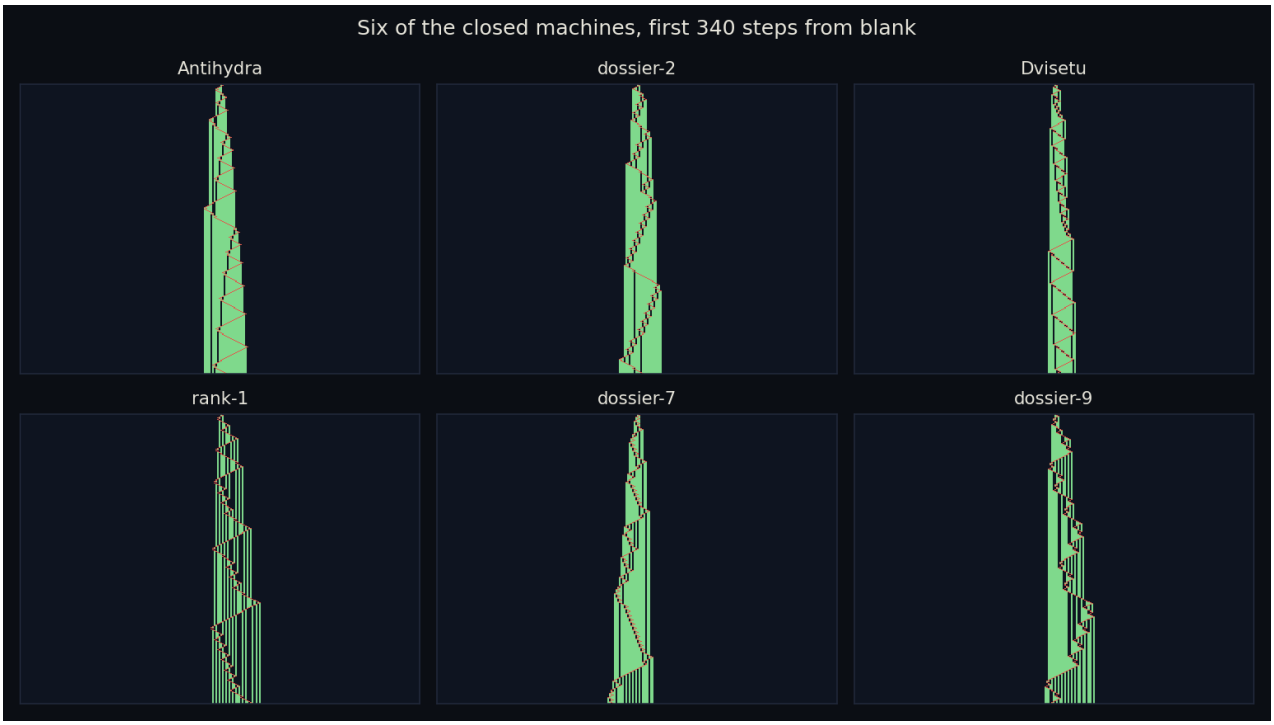
The Closed Machines of BB(6)

Nine kernel-checked conditional closures, machine by machine: what each cryptid is, how it was closed, and what remains open

Enan Srivastava (with an AI assistant; every proof checked by the Lean 4 kernel)

September 3, 2026 `github.com/Enan456/bbchallenge-proofs`

The busy beaver problem at six states is blocked by a residual set of 1,094 machines that defeat every known automated decider; the community calls them cryptids, because each hides an iterated arithmetic map of the Collatz kind. This document is a field guide to the nine that have been *closed*: for each, a machine-checked theorem states that if one explicit arithmetic conjecture holds, the machine never halts from the blank tape. Every theorem is verified by the Lean 4 kernel at the minimal axiom set $\{\text{propext}, \text{Quot.sound}\}$ with no `sorry`. All nine of the program's exact-law machines are now closed.



The method, common to all nine



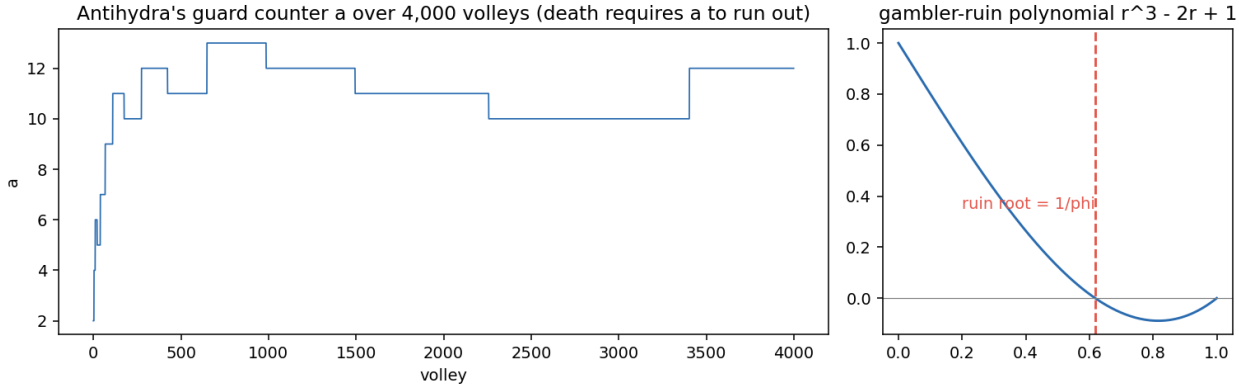
Watch. Each machine has a natural clock: a recurring configuration (a record, a spawn entry, an arrival) at which the tape has a bounded description. Five integers suffice for dossier-2/3; a triple for Antihydra; one hole and two counters for Dvisetu; a word of blocks for the rank-1 machine. **Compress.** The exact law of that description is derived empirically and fail-closed: every transition and its exact step count must be reproduced, and any miss aborts the pipeline. Laws are frozen, then validated on held-out horizons. **Certify.** The law is compiled into Lean: raw semantics; a *corridor lemma* showing every halt funnels through a single probe pattern; episode theorems with exact step counts; inductions over the unbounded middles; a concrete startup anchor; and a transport theorem showing the abstract orbit predicts the machine forever. The result, per machine: $\mathsf{Safe} \to \neg\mathsf{RawHalts}$, and (where tested) the sharpness converse: states violating the conjecture provably halt within tens of steps.

The engineering matured across the closures. The early files (dossier-2/3, 6,000–9,000 lines) prove gait-level episodes; the newest (dossiers 7/8, about 1,000 lines each) run their certified engines at single-raw-step grain, where every branch of the record law, including residue cases never observed at any empirical horizon, is simulated exactly with no assumption folded anywhere.

Antihydra

code	`1RB1RA_0LC1LE_1LD1LC_1LA0LB_1LF1RE_---0RA`
status	closed (conditional); the community's best-known cryptid
state space	hydra triple (a, m, c); tape $1^a\,0\,1^m\,0\,1^c$
keystone	`safeOrbitA_nonhalting` (2,130 lines)
conjecture	`SafeOrbitA`: the hydra orbit never dies
halt corridor	F entered only via $E0$, moving left
validation	mirror exact vs. 51.8M raw steps (10,001 records)

Antihydra iterates a hailstone-like map in which a guard counter a drifts upward on average while occasional merges spend it. Death requires the counter to run out, and the walk is a gambler's ruin with steps $-1/+2$ whose ruin probability is the root of $r^3 - 2r + 1$: exactly $1/\varphi$. The golden ratio appears in Antihydra's *hazard* where dossier-3 carries it in its *cone*: same constant, different organs. Two further facts with closed forms: the realizable dying family $(1, 2, c)$ halts in exactly $(3c^2 + 44c + 125)/4$ raw steps for odd c, and the abstract death cone grows *linearly* (branching constant exactly 1, backward merges throttled by the $a \geq 2$ guard), the third data point in the program's branching-constant series $\{\varphi,\ 2\text{-adic},\ 1\}$.



Dossier-1

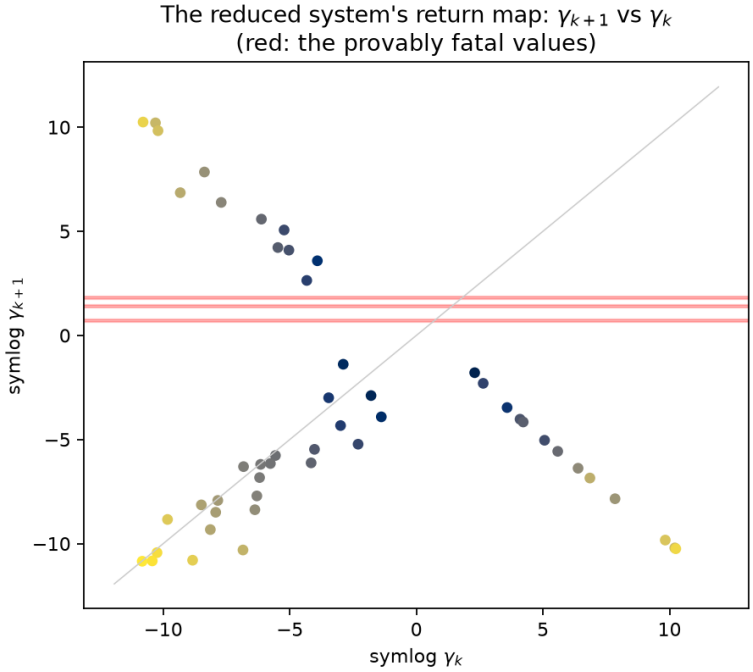
code	`1RB0LE_1LB1RC_0RF0RD_1RE0LA_1LA1RE_---0RB`
status	closed (conditional)
state space	a stack of levels (chain orbit)
keystone	`dossier1_nonhalting` (5,770 lines)
conjecture	the abstract chain orbit never dies
halt corridor	`halt_requires_c_probe_00`

The program's first original closure and the template for everything after it: the layered endgame (halt localization, then an invariant discharged by a raw-to-macro closure induction) that later files industrialized. Its 33 macro obligations were first discharged empirically at 10^8 steps, then certified.

Dossier-2 and dossier-3 (the mirror pair)

codes	`1RB1RC_1LC1LC_1RE0LD_1LB1RF_---1RA_0RD1RD` `1RB1RB_1LC0RE_---1LD_0RA1LB_1RA1LF_0LE1LE`
status	closed (conditional); one law, two machines
state space	five integers: block ends l, r, holes h_1, h_2, tag
keystones	`dossier2_nonhalting` (6,130 lines), `dossier3_nonhalting` (9,020)
conjecture	`SafeMetaOrbit`: the five-integer iteration never enters an explicit fatal set
halt corridor	the probe event: C reads 0 with a 0 to its right
validation	predictor exact over 10^9 raw steps per machine; seven episode theorems with exact step counts (crawl 10, cross 13, terminate 13, fuse 19, …)

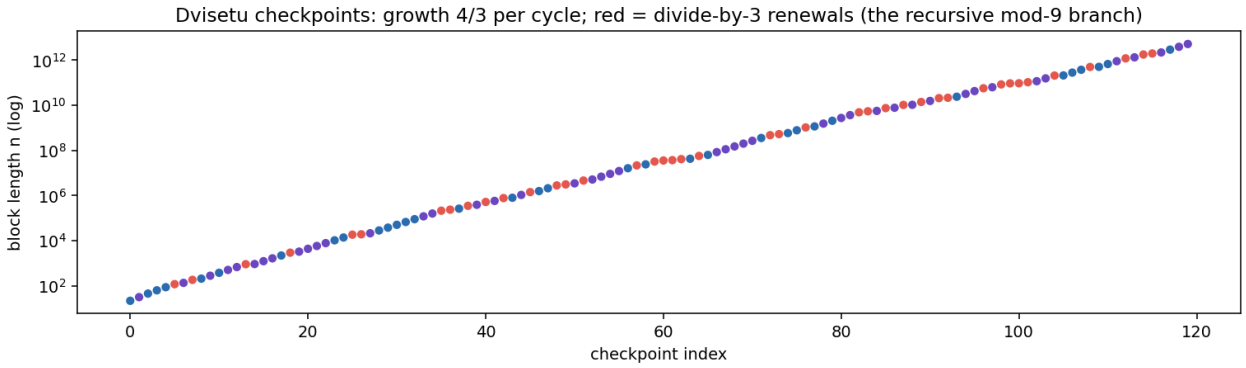
The flagship reduction. At every spawn entry the tape is a solid block of 1s with exactly two holes; seven certified episodes cover everything the machine does. Sampling at renewals collapses the dynamics to a closed-form two-counter map whose fatal set is a width-five window around the diagonal; the orbit avoids it for 10^8 iterations (about 10^{15} raw steps) with minimum margin 4. The proof routes that fail are themselves mapped: no congruence invariant to modulus 384, factor complexity $\sim 1.6^n$ (not a rotation), and 33,551 ranking-class halfplane families exhaustively refuted on the true fatal model. Dossier-3's abstract death cone is an exact Fibonacci tree ($L(k) = L(k-1) + L(k-2)$, branching at φ, mechanism: no terminate follows terminate).



Dvisetu (dossier-6)

code	`1RB0RB_1LB0LC_1RD1LC_1RE0RA_0RF1RA_---1LB`
status	closed (conditional); this program's own miner candidate (July 2026), re-surfaced blind by the deep sweep
state space	one hole, two counters: $1^a\,0\,1^b$, plus checkpoint scalars 1^n
keystone	`safeDvisetuOrbit_nonhalting` (1,942 lines)
conjecture	`SafeCheckpointOrbit` (orbit guards + the divide episode, folded as `DivideFaithful`)
halt corridor	E reading 0 with a 0 to its *right*
validation	2,746/2,746 timed record transitions at 2×10^7 steps; divide sub-rule 5/5 to $n = 18{,}648$

The simplest state geometry in the program, and a genuine generalized-Collatz specimen: drift $(a, b) \to (a{+}4, b{-}3)$ at exactly $(32b - 41)/3$ steps per record, collapse to a single block, then a renewal branched on n mod 3, whose $n \equiv 0$ branch *divides by* 3 recursively (divide while the quotient is 0 mod 9, base case by the residue mod 9). The drift gait resolved into four passes over a five-step motif, making the step count integral by construction. From dossier opening to kernel-checked closure: about 36 hours.

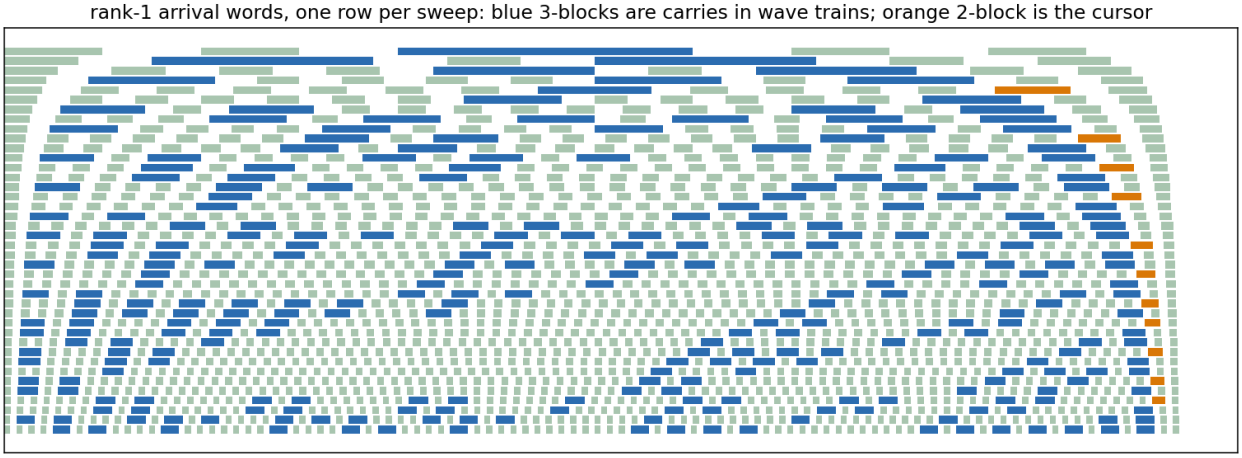


Rank-1 (dossier-5): the first growing-dimension closure

code	`1RB1RD_1LC0LC_0RD0LB_1RF0LE_0RD---_0RA1RC`
status	closed (conditional); first closure on a state space of growing dimension
state space	a *word* of 1-blocks (lengths 1/2/3, single-0 gaps) that grows without bound
keystone	`safeWordOrbit_nonhalting` (2,126 lines), zero folded assumptions
conjecture	`SafeWordOrbit`: the word orbit never dies (a single clause)
halt corridor	D reads 1 with a 1 to its left
validation	calculus frozen-exact 2,179/2,186 at 2×10^8 steps; reseed differentials exact; Lean mirror realizes every arrival configuration at the predicted cumulative time over 2.2M steps

Semantically a two-phase odometer on $\{1, 3\}$-words: the 3-blocks are carries traveling in wave trains, a deposited 2-block is a cursor that restarts the next pass at its own position, and the boundary rule is a literal block trisection $3 \mapsto 1, 1, 1$. Its death tree branches 2-adically in train lengths (the number of one-sweep preimages of a word is exactly $\prod_i 2^{\max(0,t_i-2)}$), where dossier-3's branches at φ. The certified layer works at cell grain, one level below the learned calculus, where the boundary families and the decision function are subsumed and proven.

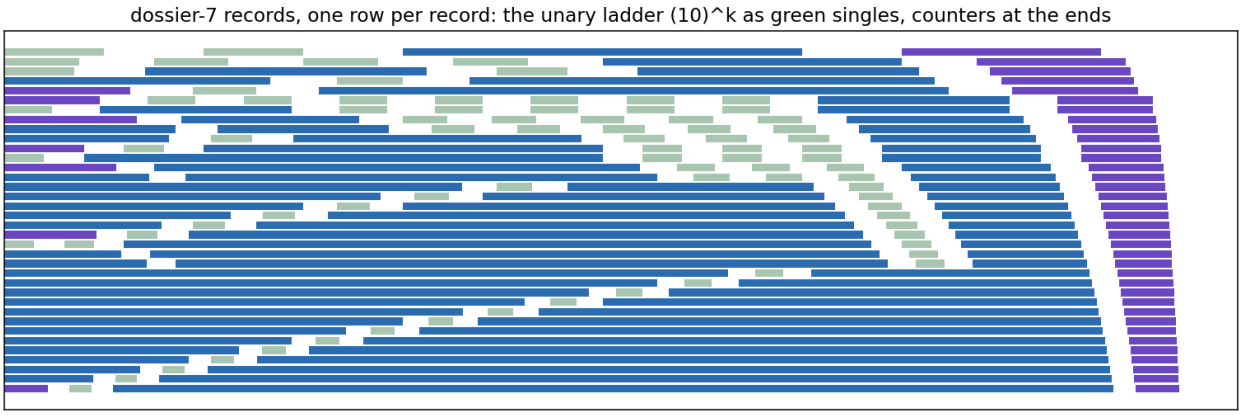
rank-1 arrival words, one row per sweep: blue 3-blocks are carries in wave trains; orange 2-block is the cursor



Dossiers 7 and 8: the raw-step-grain pair

codes	`1RB1LA_1LC0RE_1LF1LD_1RE0LC_1LB1RA_---0LA` `1RB0LC_1LC0RD_1LF1LA_1LB1RE_1RB1LE_---0LE`
status	closed (conditional); the strongest engineering of the set
state space	unary-ladder words $1^a\,0\,1\,0\,1^b\,0\,(1\,0)^k\cdots$ growing left; counters (a,b,k) and (a,b,k,g)
keystones	`safeD7Orbit_nonhalting` (1,036 lines), `safeD8Orbit_nonhalting` (1,037)
conjecture	the word orbit never dies (single clause each)
halt corridor	F entered only from $C0$; probe: C reads 0 needing a 1-guard to its left
validation	design mirrors 5,263/5,263 and 3,890/3,890 record transitions at 2×10^7 steps, validated before any Lean was written; 20,000 randomized episode plants each

Sibling machines with opposite ladder parity (and, for dossier-8, an epoch counter). Their certified engines run at single-raw-step grain: every raw move is one of three proven episodes on a boundary pair, so *every* branch of the 13- and 12-rule record laws, including residue cases whose first occurrence is projected near 3×10^{12} steps, is simulated exactly and proven faithful. Nothing is folded; the only fail-closed exits are the genuine halt patterns and fuel.



Dossier-9

code	`1RB0LF_1RC---_1LD0RF_1RA0LE_0LD1LD_0RD0RB`
status	closed (conditional); the ninth and final exact-law machine
state space	words on a right-growing frontier; parametric calculus at 100.00% held-out (271,344/271,344 segments at 8×10^6 steps) preceded the closure
keystone	`safeD9Orbit_nonhalting` (1,059 lines), single-raw-step engine, zero folds
conjecture	`SafeD9Orbit`: the word orbit never dies
halt corridor	the simplest of the set: the fatal probe *is* the halt (B reads 1; B entered only via $A0$ or $F1$)
validation	mirror 5,804/5,804 record transitions exact at 2×10^7 steps; fuel margin 2,039×; 20,000 randomized episode plants

Anatomy of a closure: the seven layers

Every closure file, from dossier-2's 6,130 lines to dossier-9's 1,059, is the same proof in seven layers. What changed across the program is only the *grain* at which layer four is built.

1. Raw semantics. The machine table as an inductive definition, one-step and n-step relations, and `RawHalts`. Nothing imported beyond Lean core: the files are self-contained down to the kernel.

2. Halt localization (the corridor lemma). Each machine has exactly one undefined transition, and working backward through the table funnels every halt through a single *probe pattern*. Verbatim from Dvisetu:

```
theorem halt_requires_e_probe_00 {c : Config} (hne : c.state <> State.F)
    (hh : RawHalts c) :
    exists d, Reaches c d /\ d.state = State.E /\
      d.tape d.head = false /\ d.tape (d.head + 1) = false
```

Everything downstream reduces to: the probe never fires. Dossier-9's corridor is the degenerate ideal, where the probe *is* the halt and safety is a one-cell predicate.

3. Probe safety as an invariant over segments. A predicate (`EProbeSafe`, `DProbeSafe`, `BProbeSafe`, per machine) plus segment-induction machinery: if every certified episode preserves safety and covers the run, the probe never fires between anchors.

4. Episode theorems: the certified choreography. The heart of each file, and where the behaviour of the machine lives. An episode theorem states: from any configuration matching a shape (given by parameters), the machine runs an *exact* number of steps, produces an *exact* output shape, and is probe-safe throughout. Verbatim from Dvisetu's drift:

```
theorem drift_step {cfg : Config} {g : Int} (n : Nat) (hn : 1 <= n)
    (hs : cfg.state = State.B) (hh : cfg.head = g + 3 * n + 2)
    (hgap : cfg.tape g = false)
    (hones : forall i, i < 3 * n + 1 -> cfg.tape (g + 1 + i) = true) : ...
```

The statement quantifies over the unbounded parameter (n, with $b = 3n + 1$), so one theorem covers infinitely many raw-step runs; the proof is an induction whose base and step are bounded evaluations. This is the compression that lets verification outrun the machine.

5. The startup anchor. From the blank tape to the first abstract state, certified step by step (235 concrete steps for dossier-2; Dvisetu's is 6; dossiers 7–9 use a *virtual anchor*, the blank tape itself realizing the empty word, so no prefix at all).

6. The abstract orbit and transport. An executable step function on the abstract state (five integers, a checkpoint scalar, a word), iterated from the anchor; a `Realizes` grammar tying abstract states to literal tapes; and the transport theorem: if the abstract step yields a successor, the raw machine, started on any realization, reaches a realization of that successor in exactly the predicted number of steps, safely. Verbatim from dossier-7's engine:

```
def wfuel (w : List Bool) : Nat := 64 * (w.length + 64) * (w.length + 64)
def wstep (st : State) (w : List Bool) : Option (State x List Bool x Nat)
-- "The sweep is faithful": whenever wstep yields a successor arrival,
-- the raw machine, started on any realization of the input, ...
```

7. The keystone. `Safe` says the abstract orbit never returns `none`; the keystone composes layers 2–6 into Safe → ¬RawHalts. The kernel then replays the whole file; the build itself prints the axiom report.

The grain ladder: three ways to build layer four

Gait grain (dossiers 1–3, Antihydra, Dvisetu): episodes are the machine's natural gaits, discovered by instrumentation (dossier-2 walks in a fixed ten-step crawl; Dvisetu's drift is four passes over a five-step motif). Maximal legibility: the certified objects are recognizably *behaviours*, and their exact step counts (10 per crawl cycle, $(32b-41)/3$ per drift record) become theorem constants. Cost: every boundary case needs its own episode, and a genuinely hard one (Dvisetu's divide renewal) must either be proven or folded into the conjecture as a named assumption.

Cell grain (rank-1): episodes are bounded-window micro-moves, one level below the learned calculus. The boundary families and even the decision function of the word calculus turn out to be *consequences* at this grain, so nothing is folded. The certified layer no longer mentions trisection or cursors by name; the conjecture absorbs the law, and the law's legibility lives in the (validated, untrusted) empirical layer.

Raw-step grain (dossiers 7–9): the terminus. Every raw move is one of three proven episodes on a boundary pair; a fail-closed sweep with quadratic fuel runs record to record. Every branch of the record law, including residue cases first occurring near 3×10^{12} steps, is simulated exactly; the only `none` exits are the genuine halt patterns and fuel. Files shrink to ∼1,000 lines. The trade: the conjecture is about an orbit the file computes but does not explain; the *explanation* (drift laws, ladders, renewals) stays in the empirical layer, which is where the mathematics of the conjecture gets done anyway.

The progression is not merely stylistic. Gait grain proves the most *about* the machine; raw-step grain proves the most *coverage* for the least trust surface. The program now holds both ends, and the right choice per machine is: gait grain where the conjecture will be attacked analytically (dossier-2/3's renewal map feeds the invariant-measure program), raw-step grain where closure and external checkability are the goal.

The trust chain, stated exactly

Trusted: the Lean kernel, propext, Quot.sound, and the machine table transcribed in layer 1. Nothing else. *Untrusted but validated*: everything that found the proofs – the finders, segmenters, calculus learners, the python mirrors, the AI that drafted the Lean. Their role is discovery; a wrong law would simply have failed to compile into theorems, and a wrong mirror is caught by the differential tests (`tests/bb6/test_*_theorems.py`: randomized episode plants checked cell-for-cell with exact step counts and probe ledgers, plus the abstract mirror replayed against millions of raw steps). The fail-closed discipline runs through every layer: learners abort on unexplained events, `wstep` returns `none` rather than guess, and the conjectures quantify over exactly the executable orbit, so there is no gap between what is conjectured and what is computed.

The certified behaviours, as a bestiary

What the nine closures actually pin down, machine by machine, is a small zoo of arithmetic behaviours:

- **Drift-and-renewal two-counter maps** (dossier-2/3 reduced form; Dvisetu at checkpoint level): a counter drains linearly while another grows, then a renewal reseeds by subtraction, swap, or division, with parity or residue choosing the branch. Death is a near-tie or a residue window; the orbits graze it (margin 4 for dossier-2/3) and survive.
- **Guarded random walks** (Antihydra): a counter with drift +0.47 per volley whose death is gambler's ruin at $1/\varphi$; deaths are early-game and confined to a bounded slab.
- **Odometers on words** (rank-1): carries as 3-blocks in wave trains, a cursor as the carry boundary, block trisection at the frontier; preimage multiplicity exactly $\prod 2^{\max(0,t_i-2)}$.
- **Unary-ladder counters** (dossiers 7–8): a $(10)^k$ ladder drained and reloaded by minor and grand renewals at growth $\sim 3/2$; the state is a fixed integer vector wearing an unbounded unary coat.
- **Lethality, measured and proven**: the closures are sharp. Danger-band states of dossier-2 halt in 6–12 raw steps when materialized; Antihydra's dying family halts in exactly $(3c^2 + 44c + 125)/4$ steps; 20,000 random hole-perturbations of live states produce zero halts. The conjectures are equivalences in practice: leaving the safe region is not unproven-safe, it is fatal.
- **The backward view**: death cones with per-machine branching constants (φ, 2-adic, exactly 1), each traceable to an alternation constraint; every record configuration provably $\geq$ 24–30 raw steps from any halt.

What the conjectures have in common, and what remains

Every closure ends at a Collatz-class orbit-avoidance statement, and the program has mapped why no shortcut exists: a sound invariant search over every learned law in the cohort (congruence closures to modulus 64, monotone functionals, two independent implementations that first cross-audited each other) found *zero* invariant-decidable machines, and ranking-class certificates were exhaustively refuted on the one system with a complete fatal model. The orbits avoid death for measure-theoretic reasons (drift, uniform hazard, Borel–Cantelli tails, seed-space density $\approx 12/X$), and finite certificates cannot see measure-theoretic truths. The standing deep project is the invariant-measure route: the renewal maps here are explicit, total, and piecewise affine with verified uniform hazard, materially better conditioned than Collatz itself.

Verify everything yourself

```
git clone https://github.com/Enan456/bbchallenge-proofs
cd bbchallenge-proofs/busy-beaver-core && lake build
# the build prints, per keystone theorem:
#   depends on axioms: [propext, Quot.sound]
```

External approval path: Palomar registration (entries pinned and validated; one browser session), bbchallenge forum publication with invited reproduction (post drafted), and DeepMath 2026 review (extended abstract at v19). No process makes these unconditional; approval means externally re-verified reductions, each machine's halting problem restated as one explicit arithmetic question.